Riley Beenders

CONTROLS ENGINEER | INDUSTRIAL AUTOMATION | RELIABILITY & TROUBLESHOOTING

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PROFESSIONAL SUMMARY

Electromechanical R&D engineer who designs, wires, commissions, and troubleshoots industrial production machines and the control systems inside them, in a manufacturing environment since May 2020. Delivers each machine from architecture through validation, commercialization, and after-sales support with minimal oversight, documents it, and performs risk analysis against EU and U.S. standards before it ships.

CORE EXPERTISE

Controls & Automation: Modbus TCP data acquisition; multi-board CAN-bus control architectures; Allen-Bradley (Rockwell) PLCs; sensor and I/O integration; industrial servo, linear-motor, and NEMA 34 stepper motion with power-failure brakes; control-system performance monitoring; tuning and calibration

Electrical Design: Electrical layout and schematic design; wiring harnesses, interconnects, and cut-panel design; 120/240 VAC machine power; independently testable electronics enclosures; engineering documentation and BOM control

Software & Scripting: Python; LabVIEW; MATLAB, Simulink, and Simscape; TypeScript and JavaScript (Electron, Node.js); PowerShell; Git and GitHub; SQLite; Linux (Debian) and Windows administration; SolidWorks, Autodesk Inventor, Siemens NX, CATIA, Fusion 360

Safety, Reliability & Leadership: Risk and hazard analysis and mitigation (2006/42/EC Machinery Directive); product safety and regulatory compliance; field troubleshooting and after-sales support; technical writing; Lean Six Sigma; hiring and technical interviews; worldwide project collaboration

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA — January 2025 - Present

Lead, Research & Development Engineer

Proteor acquired Filament Innovations in January 2025. The CTO role below continued as Lead R&D Engineer for the same machines, now the Proteor Print product line.

- Led the design and development of ICARUS-Lite, owning its mechanical, electrical, and software design, including custom AC heaters and an electrical design drawn to IEEE schematic conventions with a 120/240 VAC input; completed product changes supporting conformity with the 2006/42/EC Machinery Directive and reduced assembly time by 80% compared to previous designs.
- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.
- Built ExtrusionStudio, a plant-wide measurement and production system for three filament extrusion lines. A LabVIEW proof of concept established Modbus TCP communication with each line's tri-axial laser gauge; a from-scratch Python rewrite then replaced three single-line executables with one service that polls every line concurrently at 20 Hz, so a powered-down line never stalls the others.
- Gave ExtrusionStudio's operators, packing bench, and management their own panels: live strip charts with tolerance-band alarms and out-of-spec timers, a calibration phase that times filament from the laser to the spool so readings log onto the spool they are wound on, and a serial, CSV record, certificate, and QR label for every spool, so the plant decides spools against gauge data instead of by eye.
- Deployed a fleet of Raspberry Pi kiosks (projector, per-line touch screens, packing bench, statistics TV) that check in with the server every 30 seconds and are updated, configured, and power-switched with the work hours from one settings panel.
- Built BulkSSH (Electron, TypeScript, open source at github.com/RileyBeenders/BulkSSH), a Windows app that opens many SSH devices in a grid of live terminals, mirrors keystrokes, and broadcasts commands with per-device exit codes, so every change to the Pi image is tested on every Pi in one pass instead of one SSH window at a time.
- Standardized POSEIDON and ICARUS on one linear-motor architecture with swappable mounts, replacing servo-driven ballscrews to cut product-to-product architectural complexity and simplify service; validated 4-micron precision at 4 m/s with a 30 kg moving mass.
- Implemented ProteorPrint OS with SimplyPrint, enabling remote operation, live video monitoring, AI-assisted failure detection and alerts, and material, waste, and outcome metrics for analytics-driven operational decisions.
- Provide cross-functional technical support for the product line, from troubleshooting to after-sales service; co-developed a custom hotend with an external manufacturing partner, improving reliability and reducing cost for the company's materials.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.

Filament Innovations | Whitehall, PA (acquired by Proteor, January 2025) May 2020 - January 2025

Chief Technology Officer January 2023 - January 2025

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while retaining hands-on responsibility for mechanical, electrical, firmware, and software systems.
- Led the design and launch of Gen3 POSEIDON, combining a 1 m^3 build volume, filament-and-pellet dual extrusion, braked NEMA 34 Z drives that prevent uncontrolled motion during power loss, industrial-servo XY motion, and a custom gantry.
- Researched high-speed linear motors that moved a 30 kg payload at 3 m/s with 5-micron accuracy and conducted early 6-axis robotic non-planar printing research.

Lead Electro-Mechanical Engineer June 2022 - January 2023

- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging on larger, modular machines.
- Developed independently testable electronics enclosures that could be installed in minutes and coordinated custom harnesses, panel designs, and hardware improvements with vendors.
- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.

Industrial Engineer April 2021 - June 2022

- Developed next-generation ICARUS and POSEIDON models with higher-performance components and expanded motion safety through NEMA 34 motors, planetary gearboxes, and power-failure brakes.
- Designed THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry printer for a client producing the world's largest 3D-printed statue, and managed CAD, BOM, sourcing, fabrication, assembly, firmware tuning, support, and training.

Engineering Technician May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized print parameters while documenting design limitations and improvement opportunities.

ADDITIONAL RELEVANT EXPERIENCE

Blue Mountain Ski Resort - Ski Lift Operator | November 2019 - January 2021 | Performed motor and control safety checks on passenger ski lifts and supported the maintenance team during lift breakdowns; operated the lifts for thousands of guests.

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed more than 750 wireless access points across six school buildings; accelerated deployment of more than 2,500 Chromebooks with an Arduino-based tool that automated setup-data entry.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and operate a small online store for printer heat inserts and accessories.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering December 2024

MATLAB, Simulink, Simscape, and Machine Learning Onramps, MathWorks 2026

Lean Six Sigma Yellow Belt and White Belt, Educate 360 2025

Certificate in Engineering Design and Tools, Penn State University 2023

Mechanical Engineering Certificate, Lafayette College 2019